

NeuralRetail AI-Powered Retail Analytics Platform

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Contents

1	Abstract	2
2	Introduction	2
3	Project Overview	2
4	Data Collection and Preprocessing	2
4.1	Dataset Sources	3
4.2	Data Cleaning and Preprocessing	3
4.3	Feature Engineering	3
5	Customer Segmentation	4
6	System Architecture Design	5
7	Dashboard Design and Business Intelligence Modules	6
7.1	Executive Overview	6
7.2	Sales Analytics	6
7.3	Product Analytics	6
7.4	Customer Analytics	6
7.5	Geographic Analytics	7
7.6	Forecasting and AI	7
7.7	Operations Dashboard	7
7.8	Recommendation Engine	7
8	Deployment	7
9	Conclusion	8

1 Abstract

NeuralRetail is an AI-powered retail analytics platform developed to solve critical business challenges in customer intelligence, demand forecasting, inventory optimization, churn prediction, and product recommendation. The system combines machine learning models, business analytics, and interactive dashboards to generate actionable insights for retail organizations.

The platform utilizes the Online Retail II dataset for customer analytics and the M5 Forecasting dataset for sales forecasting and inventory optimization. Multiple machine learning techniques including K-Means Clustering, Gaussian Mixture Models, Logistic Regression, Random Forest, XGBoost, and collaborative filtering were evaluated and implemented.

The final solution was deployed using Streamlit Cloud, providing a live analytics dashboard capable of supporting data-driven decision-making for modern retail businesses.

2 Introduction

The retail industry generates large volumes of customer, sales, and inventory data every day. Transforming this data into actionable insights is essential for improving customer satisfaction, increasing revenue, and optimizing business operations. However, traditional reporting systems often fail to provide predictive insights required for effective decision-making.

NeuralRetail is an AI-powered retail analytics platform developed to address key retail challenges including customer segmentation, churn prediction, demand forecasting, inventory optimization, and product recommendation. By combining machine learning techniques with interactive business intelligence dashboards, the platform enables data-driven decision-making and operational efficiency.

3 Project Overview

NeuralRetail is an end-to-end retail analytics solution built using the Online Retail II and M5 Forecasting datasets. The platform integrates multiple analytical modules including customer segmentation, churn prediction, demand forecasting, inventory optimization, recommendation systems, and business intelligence dashboards.

The system utilizes machine learning algorithms such as K-Means Clustering, Random Forest, Logistic Regression, and XGBoost to generate business insights. An interactive Streamlit dashboard was developed to visualize key performance indicators, customer behavior, sales trends, forecasting results, and inventory intelligence. The final solution was deployed on Streamlit Cloud, providing an accessible web-based analytics platform for retail decision-making.

4 Data Collection and Preprocessing

The NeuralRetail platform utilizes two publicly available datasets: the Online Retail II dataset and the M5 Forecasting Accuracy dataset. These datasets were selected to support customer intelligence, sales forecasting, inventory optimization, and recommendation system development.

4.1 Dataset Sources

Online Retail II Dataset

Source: <https://archive.ics.uci.edu/dataset/502/online+retail+ii>

The Online Retail II dataset contains transactional records from a UK-based online retail company. The dataset includes customer purchase histories, product information, transaction timestamps, quantities purchased, pricing information, and geographic details. This dataset was primarily utilized for customer analytics, customer segmentation, churn prediction, recommendation system development, and business intelligence reporting.

M5 Forecasting Accuracy Dataset

Source: <https://www.kaggle.com/competitions/m5-forecasting-accuracy>

The M5 Forecasting dataset contains large-scale retail sales records collected across multiple stores and product categories. The dataset includes historical sales information, calendar events, product metadata, and store-level information. This dataset was utilized for demand forecasting, inventory optimization, sales trend analysis, and operational planning.

4.2 Data Cleaning and Preprocessing

Raw retail datasets often contain missing values, duplicate transactions, cancelled orders, and inconsistent records. Therefore, extensive preprocessing was performed before model development.

The following data cleaning operations were applied:

- Removed duplicate transaction records.
- Removed rows containing missing Customer IDs.
- Removed cancelled transactions and invalid invoices.
- Filtered records containing negative quantities and incorrect pricing values.
- Converted date columns into datetime format for time-series analysis.
- Generated revenue values using $\text{Quantity} \times \text{Unit Price}$.
- Standardized customer and product information.
- Handled missing and inconsistent categorical values.

4.3 Feature Engineering

Feature engineering was performed to transform raw transactional data into meaningful business intelligence features.

For customer analytics, Recency, Frequency, and Monetary (RFM) features were generated:

- **Recency:** Number of days since the customer's last purchase.
- **Frequency:** Total number of transactions completed by a customer.

- **Monetary:** Total revenue generated by a customer.

For churn prediction, additional features were engineered:

- Average Order Value
- Frequency Score
- Monetary Score
- Customer Segment Labels

For demand forecasting, time-series features were generated:

- Day of Week
- Month
- Week Number
- Lag 7 Sales
- Lag 28 Sales
- Rolling Mean (7 Days)
- Rolling Mean (28 Days)
- Event Indicators
- Seasonal Features

The processed datasets were subsequently stored as structured CSV files and utilized throughout the machine learning pipeline for training, evaluation, and dashboard visualization.

5 Customer Segmentation

Customer segmentation was performed using the Recency-Frequency-Monetary (RFM) framework. RFM analysis is a widely adopted customer analytics technique that measures how recently a customer purchased, how frequently purchases occur, and how much revenue is generated.

Recency was calculated as the number of days since the customer's most recent transaction. Frequency represented the total number of purchase transactions completed by each customer. Monetary value represented the cumulative revenue generated by the customer.

After feature extraction, all RFM variables were standardized using StandardScaler to ensure equal contribution during clustering.

Three clustering algorithms were evaluated:

- K-Means Clustering
- Density-Based Spatial Clustering (DBSCAN)

- Gaussian Mixture Models (GMM)

K-Means achieved the highest silhouette score of 0.61 and was selected as the final segmentation model. DBSCAN was used to identify high-value outlier customers, while GMM was evaluated as a probabilistic clustering alternative.

The final segmentation identified multiple customer personas, including Loyal Customers, High-Value Customers, VIP Customers, and At-Risk Customers. These segments provide actionable business insights for targeted marketing campaigns and customer retention strategies.

6 System Architecture Design

The NeuralRetail platform follows a modular architecture that integrates data processing, machine learning, business analytics, and dashboard visualization into a unified workflow. The system is designed to transform raw retail transaction data into actionable business insights through a series of analytical modules.

The overall architecture consists of the following components:

1. Data Collection Layer

Retail transaction data was collected from the Online Retail II dataset, while sales forecasting data was obtained from the M5 Forecasting dataset.

2. Data Preprocessing Layer

Raw datasets were cleaned, transformed, and validated. Missing values, duplicate records, cancelled transactions, and invalid entries were removed. Feature engineering techniques were applied to generate business-specific attributes.

3. Analytics and Machine Learning Layer

This layer contains multiple machine learning modules including customer segmentation, churn prediction, demand forecasting, inventory optimization, and recommendation systems.

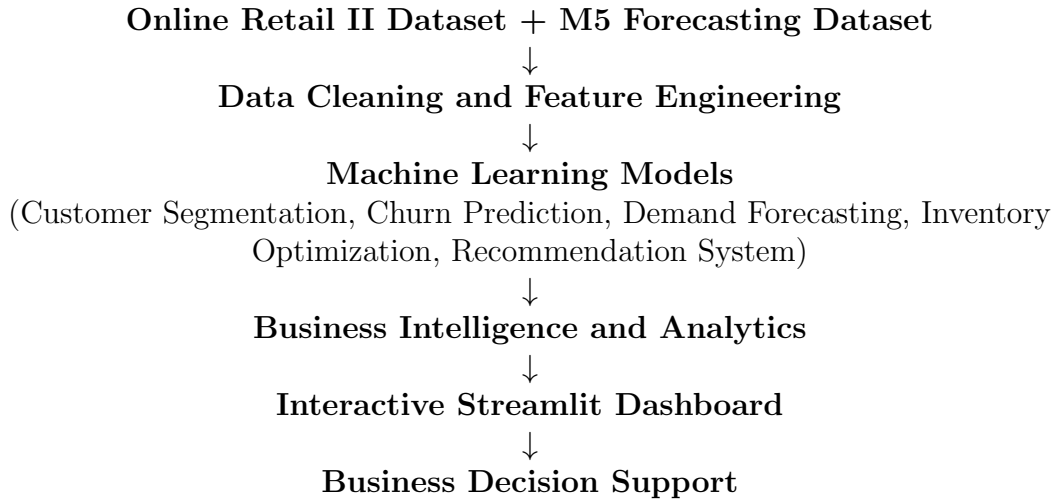
4. Business Intelligence Layer

Analytical results generated by machine learning models are transformed into meaningful business metrics, key performance indicators (KPIs), and operational insights.

5. Visualization and Deployment Layer

A Streamlit-based dashboard was developed to provide interactive visualization of business analytics, forecasting results, customer insights, and inventory intelligence. The final application was deployed using Streamlit Cloud.

The architecture enables seamless integration between data processing, predictive analytics, and business intelligence, allowing stakeholders to monitor business performance and support data-driven decision making.



7 Dashboard Design and Business Intelligence Modules

A multi-page Streamlit dashboard was developed to provide interactive business intelligence capabilities.

7.1 Executive Overview

The Executive Overview page provides a consolidated summary of key business performance indicators. Metrics including total revenue, total customers, total products, total orders, average daily demand, churn rate, and forecasting accuracy are displayed through KPI cards. This page serves as the primary management dashboard for executive decision-making.

7.2 Sales Analytics

The Sales Analytics module visualizes business sales trends over time. Monthly revenue trends, weekday revenue patterns, and order volume distributions are presented through interactive charts. Export functionality enables stakeholders to download analytical reports for further investigation.

7.3 Product Analytics

The Product Analytics page evaluates product performance across the business portfolio. Top-selling products, highest revenue-generating products, and product demand patterns are visualized through ranking tables and interactive charts. This module assists inventory managers in identifying profitable products and demand trends.

7.4 Customer Analytics

The Customer Analytics module presents customer segmentation results and churn analysis. Segment distributions, customer behavioral patterns, and retention indicators are displayed through visual dashboards. This module supports customer relationship management and targeted marketing initiatives.

7.5 Geographic Analytics

Geographic Analytics provides country-level and region-level revenue analysis. Revenue contribution from different geographic markets is visualized to identify high-performing regions and potential growth opportunities.

7.6 Forecasting and AI

This module presents machine learning forecasting results generated using the XGBoost forecasting model. Historical demand data and predicted future demand values are displayed together, allowing stakeholders to assess forecasting accuracy and business trends.

7.7 Operations Dashboard

The Operations Dashboard integrates inventory optimization outputs, including Economic Order Quantity (EOQ), Safety Stock, Reorder Point calculations, and ABC-XYZ inventory classifications. These metrics assist operational teams in reducing stockouts and minimizing inventory holding costs.

7.8 Recommendation Engine

The Recommendation Engine was developed to enhance customer experience and support product discovery within the retail ecosystem. The system utilizes item-based collaborative filtering techniques to identify relationships between products based on historical customer purchasing behavior.

A product similarity matrix was generated using customer-product interaction data, enabling the system to recommend products that are frequently purchased together or exhibit similar purchasing patterns. These recommendations can be used to improve cross-selling and upselling opportunities, increase average order value, and enhance customer engagement.

The recommendation module was integrated into the NeuralRetail dashboard, allowing users to retrieve relevant product suggestions through an interactive interface. Although the recommendation engine was developed and tested successfully, the cloud deployment version was optimized to reduce storage requirements and improve deployment efficiency.

The recommendation system demonstrates how machine learning can be utilized to deliver personalized shopping experiences and support data-driven retail marketing strategies.

8 Deployment

The NeuralRetail platform was deployed using Streamlit Community Cloud to provide a publicly accessible analytics dashboard.

Deployment Platform:

- Streamlit Community Cloud
- Python 3.10

- GitHub Integration

Application URL:

<https://neuralretail-mw.streamlit.app/>

Source Code Repository:

<https://github.com/MeAbhishek09/NeuralRetail>

The deployment enables stakeholders to access forecasting results, customer intelligence insights, inventory optimization metrics, and business analytics through a web-based interface without requiring local software installation.

9 Conclusion

This project presented the design and development of NeuralRetail, an AI-powered retail analytics platform that integrates customer intelligence, demand forecasting, inventory optimization, churn prediction, recommendation systems, and business intelligence dashboards into a unified analytical framework.

The platform successfully utilized the Online Retail II and M5 Forecasting datasets to generate actionable business insights through machine learning and data analytics techniques. Multiple models and analytical approaches were evaluated, including K-Means clustering, DBSCAN, Gaussian Mixture Models, Logistic Regression, Random Forest, and XGBoost.

Customer segmentation enabled the identification of distinct customer groups, while churn prediction helped identify customers at risk of leaving. Demand forecasting models provided accurate future sales estimates, supporting inventory planning and operational decision-making. Inventory optimization techniques including EOQ, Safety Stock, Re-order Point calculations, and ABC-XYZ analysis improved inventory intelligence and stock management capabilities.

An interactive Streamlit dashboard was developed to visualize key business metrics, customer behavior, forecasting results, geographic insights, and operational analytics. The final solution was successfully deployed on Streamlit Cloud, making the platform accessible through a web-based interface.

Overall, NeuralRetail demonstrates the practical application of Artificial Intelligence, Machine Learning, and Business Intelligence techniques in solving real-world retail challenges. The project highlights the potential of data-driven decision-making in improving customer engagement, operational efficiency, inventory management, and business performance.